

Quenton Ni

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EDUCATION

University of Minnesota, College of Science and Engineering

M.S. in Computer Science | GPA: 4.00

Minneapolis, MN

Expected May 2027

University of Minnesota, College of Science and Engineering

B.S. in Computer Science, Minor in Management Information Systems | GPA: 4.00 | High Distinction

Minneapolis, MN

May 2026

EXPERIENCE

IT Intern (Applied AI & Data Automation)

Land O'Lakes Inc.

Arden Hills, MN

May 2026 – July 2026

- Automated an estimated 120 hours of sales research by building an Azure AI Foundry web agent that scraped and normalized K-12 meal data from 100+ districts into Snowflake.
- Fine-tuned and evaluated RF-DETR and YOLO object detection models to flag pallet defects for manual review in a claims process associated with approximately \$3M in annual losses.

Software Engineer (Capstone)

Medtronic

Minneapolis, MN

Sep. 2025 – May 2026

- Built and optimized a FastAPI/React workflow that matched diagnostic logs to Azure DevOps bugs using cached checksums, WIQL, and concurrent retrieval, enabling engineers to confirm, revise, or dismiss suggested matches.
- Integrated automated unit tests and SonarQube analysis to Azure DevOps pipelines while coordinating delivery across four student teams and a research lab.

Research Assistant | NSF REU & UROP Grant Recipient

University of Minnesota-Twin Cities

Minneapolis, MN

Sep. 2025 – May 2026

- Redesigned a ventricular arrhythmia classification model using depthwise separable convolutions, reducing its parameter count from 1,534 to 390.
- Implemented quantization-aware training with INT4 precision for ventricular arrhythmia classification, contributing to a 31× reduction in theoretical model footprint while retaining an F-beta score of 0.956 versus 0.964 for the FP32 baseline.

Computer Science/Machine Learning Teaching Assistant

University of Minnesota-Twin Cities | Data Science and AI Hub

Minneapolis, MN

Sep. 2024 – May 2026

- Led weekly labs and office hours covering Java, C systems programming, and x86-64 assembly, supporting 400+ students and testing course materials.
- Supported PhD-led instruction for 90+ high school seniors, teaching machine learning concepts including deep neural networks, transformers, and reinforcement learning.

PROJECTS

Multi-Dialect Vietnamese Text-to-Speech

- Fine-tuned and evaluated a multi-dialect Vietnamese TTS system using F5-TTS on approximately 71 hours of dialect-annotated speech across Northern, Central, and Southern Vietnamese.
- Built and presented an interactive Gradio demo for dialect-controlled Vietnamese speech synthesis with generated audio playback and mel-spectrogram visualization.

Gamblr

- Engineered a real-time tracking application using Vue.js and Firebase Firestore, featuring a customizable drag-and-drop dashboard for live spending analysis and habit tracking.
- Enhanced user interaction by integrating Google Cloud Speech-to-Text via Firebase Functions and rendering an interactive 3D slot machine simulation using Three.js.

Distributed Auction Marketplace System

- Built a distributed auction marketplace with a centralized controller, stateless service nodes, and replicated SQLite storage nodes orchestrated through Docker Compose.
- Implemented controller-managed health checks, automatic primary-backup failover, and request-rate autoscaling with round-robin routing to maintain availability during node failures.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL

Machine Learning & Data: PyTorch, NumPy, pandas, scikit-learn, Hugging Face Transformers and Datasets, RAG, multi-modal embeddings, Weights & Biases, Gradio, model quantization

Software & Platforms: FastAPI, React, Vue.js, Docker, Azure AI Foundry, Azure DevOps, Firebase/Firestore, PostgreSQL, SQLite, Snowflake, Git/GitHub, SonarQube

Relevant Coursework: Machine Learning Fundamentals, Applied Machine Learning, Natural Language Processing, Distributed Systems, Operating Systems, Computer Vision, Machine Learning for Healthcare, Cloud Computing, Computer Networks